Fault Detection Classification and Location of Transmission Line Using AI Techniques

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Abstract— This work presents an intelligent protection scheme in high voltage transmission line using artificial neural network (ANN). The objective is to analyse the performance of artificial neural network based relay to detect, classify and locate faults in transmission line using feed-forward network along with backpropagation algorithm. The input features of the neural network are fundamental frequency voltage and current magnitudes extracted by discrete- Fourier transform. One full cycle DFT has been used extract RMS of fundamental frequency components of voltage and current signals. These samples are then normalized between range of 0 to 1 as a ratio of post fault to pre- fault values of voltage and current signal so as to suit input of ANN. The inputs will be applied to ANN pattern recognizer. The artificial neural network based relay is very effective to locate faults on transmission lines and achieve satisfactory performances

Key words: Fault Detection Classification, Transmission Line, AI Techniques

I. INTRODUCTION

The global electricity system has expanded rapidly over the last few decades, necessitating the installation of a massive number of additional transmission lines. Furthermore, the requirement for a consistent and dependable supply of electricity to end customers has grown as a result of the introduction of new marketing concepts like deregulation [1]. A power system malfunction is one of the most significant elements that prevents the constant flow of energy and power. A fault in the power system is any aberrant current flow in any component of the power system. These flaws cannot be totally prevented since some of them also result from natural causes that are outside of human control [2]. Therefore, it is critical to have a well-coordinated protection system that can identify the type of fault, precisely determine its location within the power system, and detect any irregular current flow in the power system [3]. Devices that identify faults and ultimately isolate the affected area from the rest of the power system are often in charge of handling them [4].

Therefore, identifying, categorizing, and locating faults provide some significant obstacles for the continuous delivery of electricity [5]. The High Voltage Transmission Lines (that transmit the power generated at the generating plant to the high voltage substations) are more prone to the occurrence of a fault than the local distribution lines (that transmit the power from the substation to the commercial and residential customers) because there is no insulation around the transmission line cables unlike the distribution lines [6]. The reason for the occurrence of a fault on a transmission line can be due to several reasons such as a momentary tree contact, a bird or an animal contact or due to other natural reasons such as thunderstorms or lightning. Most of the research done in the field of protective relaying of power systems concentrates on transmission line fault protection due to the fact that transmission lines are relatively very long and can run through various geographical terrain and hence it can take anything from a few minutes to several hours to physically check the line for faults[7]. The quicker we restore power, the more the automatic fault location can improve the system's reliability. the more money and valuable time we save. In order to make it easier to locate these defects, several utilities are including fault locating equipment into their power quality monitoring systems that are outfitted with global information systems [8]. The following categories essentially describe fault location techniques:

1) Impedance measurement based methods
2) Travelling-wave phenomenon based methods
3) High-frequency components of currents and voltages generated by faults based methods
4) Intelligence based method

II. FAULTS IN POWER SYSTEM

There are two types of Faults in Power System. From that Symmetrical faults and Unsymmetrical Faults are there. The Symmetrical faults also classified into two faults. One is three line-to-ground (LLLG) fault and another is Line-to-line-to-line (LLL) fault.

A. Symmetrical Faults

The fault that results in symmetrical fault current (equal currents with 120 displacement) is known as symmetrical fault. Three phase faults is an example of symmetrical fault where all three phases are short circuited with or without involving the ground.

These faults rarely occur in practice as compared with unsymmetrical faults. Two kinds of symmetrical faults include line to line to line (L-L-L) and line to line to line to ground (L-L-L-G) as shown in figure below.

A rough occurrence of symmetrical faults is in the range of 2 to 5% of the total system faults. However, if these faults occur, they cause a very severe damage to the equipment’s even though the system remains in balanced condition.

The analysis of these faults is required for selecting the rupturing capacity of the circuit breakers, choosing set-phase relays and other protective switchgear. These faults are analysed on per phase basis using bus impedance matrix or Thevenin's Theorem.

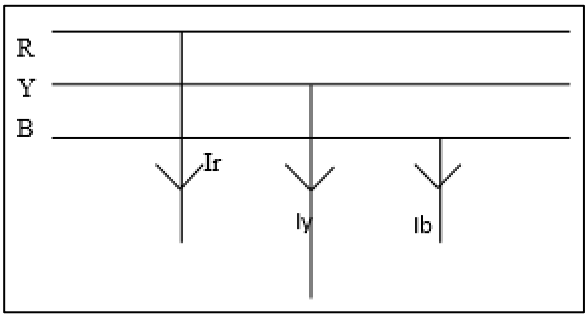


Fig. 1: Three line-to-ground fault

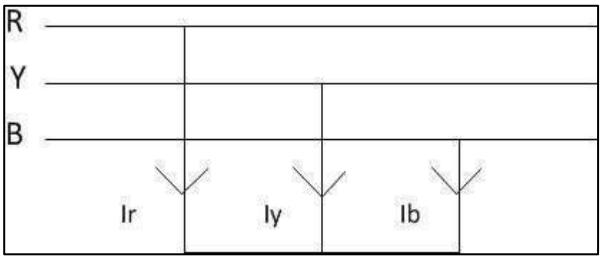


Fig. 2: Line-to-line-to-line (LLL)

B. Unsymmetrical Fault

Unsymmetrical faults are the faults that happen in the power system network most frequently. Unsymmetrical fault currents (varying in size with uneven phase displacement) result from this type of fault. Because they result in unbalanced currents in the system, these faults are also known as unbalanced faults.

There are various types of unsymmetrical faults occur in power systems are: Single Line-to-Ground (LG) fault, Line-to-Line (LL) fault, Double Line-to-Ground (LLG) fault

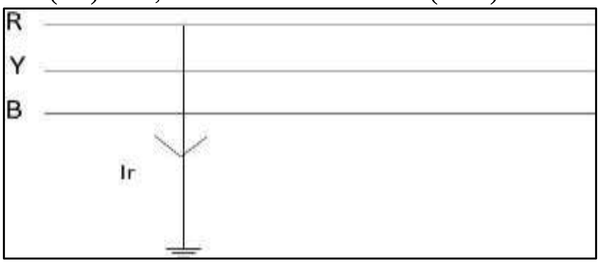


Fig. 3: Single line-to-ground (LG) fault

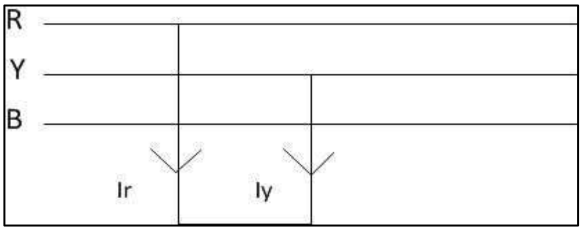


Fig. 4: Line-to-line (LL) fault

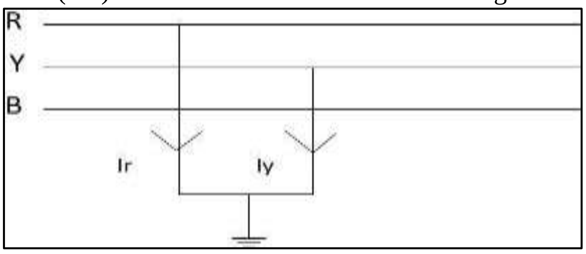


Fig. 5: Double line-to-ground (LLG) fault

III. POWER PROTECTION SYSTEMS

One of the most important components of a power protection system is the relay which is a device that trips the circuit breakers when the input voltage and current signals correspond to the fault conditions designed for the relay operation [10]. Relays in general can be classified into the following categories

1) Directional Relays: These relays respond to the phase angle difference between two inputs to the relay.
2) Differential Relays: These relays respond to the magnitude of the algebraic sum of two or more of its inputs.
3) Magnitude Relays: These relays respond to the magnitude of the input quantity.
4) Pilot Relays: These relays respond to the input that are communicated to the relay from a remote location.
5) Distance Relays: These relays respond to the ratio of two input phasor signals.
6) Travelling wave based relay: These relay respond to front of travelling wave of voltage and current signals.
7) Intelligent techniques: These are intelligent relay which have more accuracy than conventional ones.

A. Transmission line fault location techniques

The transmission line fault location process, as mentioned before, has been researched for a while and several innovative and efficient techniques have been proposed and analyzed by several authors. These techniques can be broadly classified as Impedance based methods, Travelling wave based methods and Artificial Intelligence based methods. Each of these methods is discussed briefly in the following subsections.

B. Fault Detection and Classification

Designing Fault detection and classification is very important in distance relay to avoid unnecessary tripping of circuit breakers. Appropriate selection for fault types will minimize errors such as in cases where a tripping of three phase results when only one phase clearance is required. Also, it is necessary to calculate the apparent impedance because every types of fault has own impedance algorithm.

The following steps are used in the design of a subsystem:

– Measuring the current for each phase at relay location using Three- Phase V-I Measurement block.
– Comparing the fault current with pre-fault current for each phase using Ifand If Action subsystems. The output of the comparator will be one at the time that the current exceeds the pre-fault current (reference current)

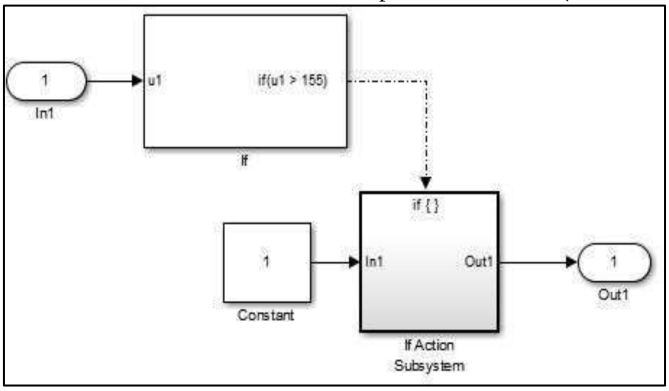


Fig. 6: Fault comparator structure

C. Neural Networks Based Methods

Neural networks have been put in use for fault location quite recently and have gained significant importance since Sobajic and Pao used neural networksfor the prediction of critical clearing time. Wide usage of neural networks started by late eighties and during early nineties. Neural networks are usually used to achieve greater efficiency in fault detection, classification and location. A lot of research has been done and abundant literature has been published in the field of fault location using neural networks. Certain significant techniquesand results that have been published are briefly discussed here. A majority of the work mentioned here made use of feed-forward multilayer perceptron technique.

Neural networks were utilized by Kulicke and Dalstein to identify faults on transmission lines and distinguish between arcing and nonarcing problems. Rikalo, Sobajic, and Kezunovic have presented a new method for the use of neural networks in the identification and localization of high speed errors[11]. Chen and Maun have extensively studied neural network-based single-ended fault location methods, whereas Song employed neural networks for series compensated line fault location.

IV. RESULTS

Ia	Ib	Ic	Ig	AG	BG	CG	AB	AC	BC	ABG	ABCG
1	0	0	1	1	0	0	0	0	0	0	0
0	1	0	1	0	1	0	0	0	0	0	0
0	0	1	1	0	0	1	0	0	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0	1	0	0	0
0	1	1	0	0	0	0	0	0	1	0	0
1	1	0	1	0	0	0	0	0	0	1	0
1	0	1	1	0	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	1

Table 1: The logic for classification of various type of faults

AG	$I_A \overline{I_B}\,\overline{I_C} I_G$	ABG	$I_A I_B \overline{I_C} I_G$
BG	$I_A \overline{I_B}\,\overline{I_C} I_G$	ACG	$I_A \overline{I_B} I_C I_G$
CG	$I_A \overline{I_B}\,\overline{I_C} I_G$	BCG	$I_A \overline{I_B} I_C I_G$
AB	$I_A \overline{I_B}\,\overline{I_C} I_G$	ABC	$I_A I_B \overline{I_C} I_G$
AC	$I_A \overline{I_B}\,\overline{I_C} I_G$	ABCG	$I_A I_B I_C I_G$
BC	$I_A \overline{I_B}\,\overline{I_C} I_G$		

Table 2: Identification of Faults

Table 1 and 2 shows the fault detection and classification subsystem. The inputs of this subsystem are phase currents and the outputs are the signalswhich will determine which impedance algorithm for the apparent impedance calculation for each type of possible fault. Table 1 presents the methods usedwith each fault type.

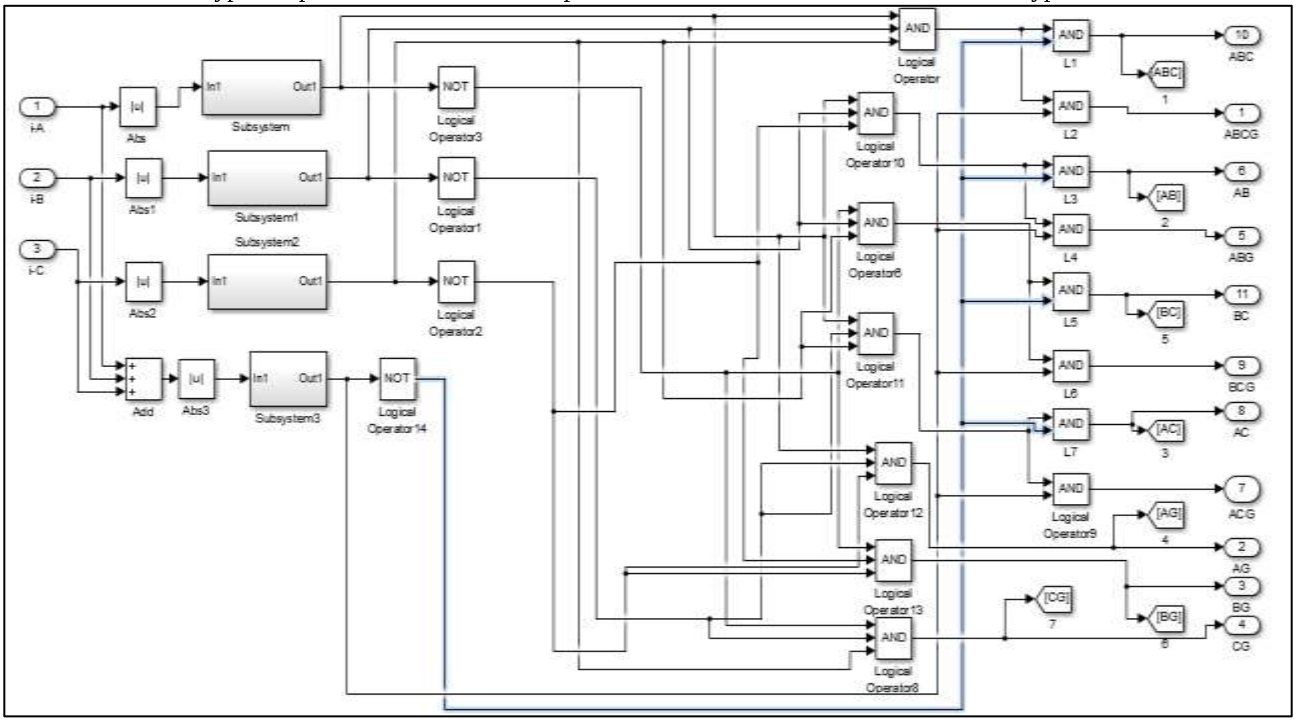


Fig. 7: Fault detection and classification block

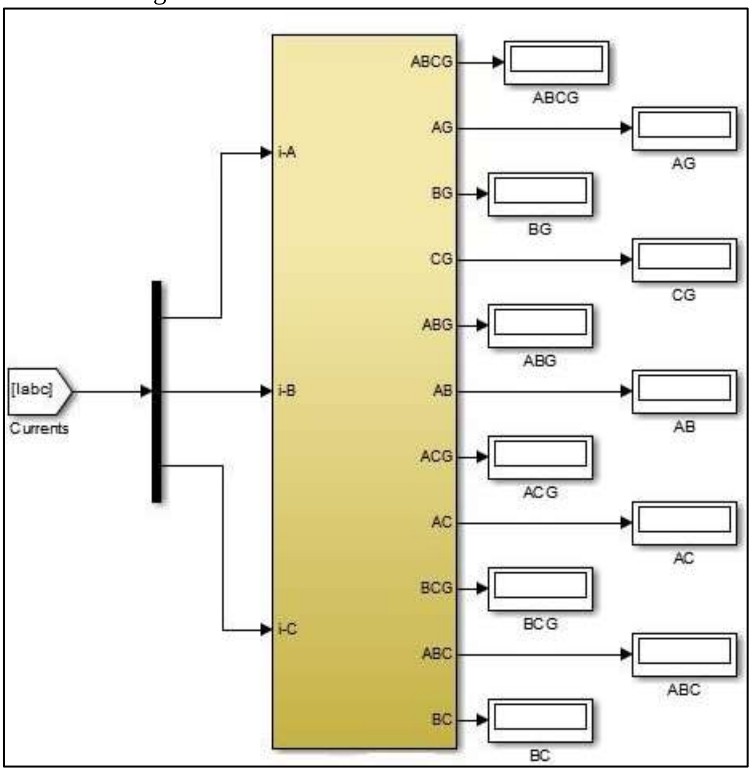


Fig. 8: Overall fault detection and classification block

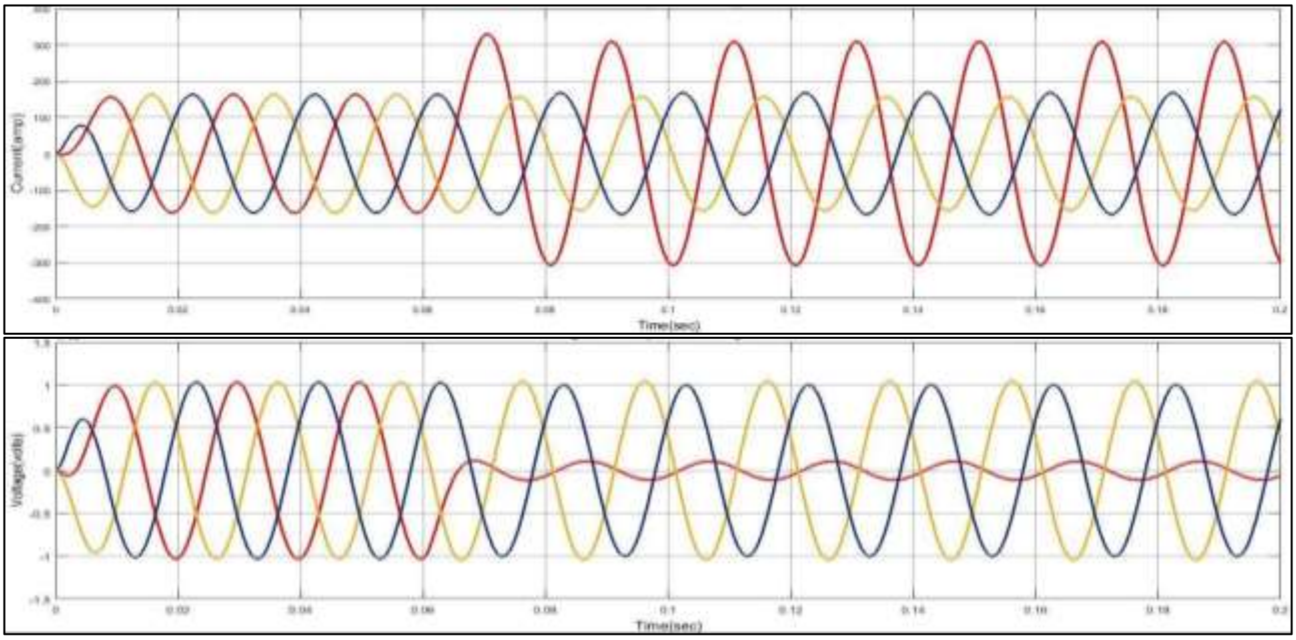
A. Apparent Impedance Measurement

Apparent impedance measurement block includes all impedance algorithms to calculate the apparent impedance by measurement the voltages and currents at relay location. The output signal from fault detection and classification block select which impedance algorithm must be used.

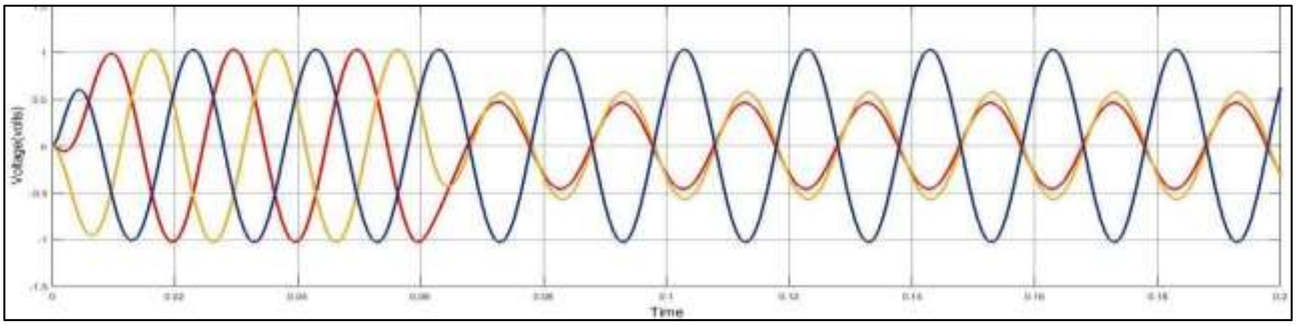
V. WAVEFORMS

The transmission lines were simulated for various faults and the graph of respective faults are shown below.

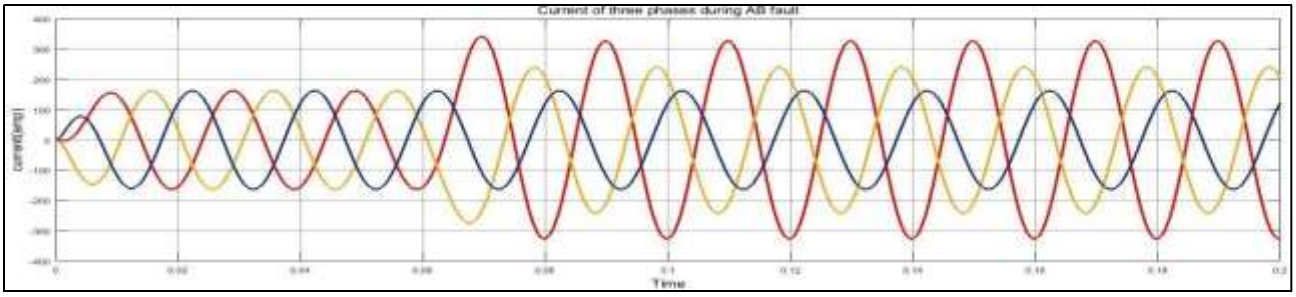
A. Waveform of current signal of three phases during AG fault



B. Waveform of voltage signal of three phases during AG fault



C. Waveform of voltage signal of three phases during AB fault



D. Waveform of Current Signal of Three Phases during AB Fault

The impedance calculated by simulating distance relay in order to identifyzone of fault is in given table.

Sr No	Faulttype	Fault location (km)	Fault resistance (Ω)	Fault inceptionangle	Impedance (Ω)
1	AG	20	0.001	0	5.29+j2.30
2	AG	40	0.001	0	9.98+j4.59
3	AG	60	0.001	0	16.01+j8.89
4	AG	80	0.001	0	21.01+j9.18

Table 3: Impedance Measured for Different fault location

VI. CONCLUSIONS

The conventional methods like distance relay, travelling based relay are good but they have its own limitations. The accuracy of these methods varies for system to system. Distance relays are subjected to difficulties such as fault resistance variation, over reach, under reach, power swing and remote in-feed. In travelling wave based methods there is need of fault transient detectors, diagnostic software and it is difficult to test. Artificial Neural network approach can be applied to protection of transmission lines over conventional methods. It can overcome most of difficulties mentioned in above points. A network can be developed by training it with past data and can be tested.

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